

Aspergillus fumigatus in the hospital environment in Belgium: investigation of azole resistance

N. Motteu¹ • H. Debergh^{1,2} • R. De Paepe¹ • B. Goemaere¹ • A. Packeu^{1,2}

1. Mycology and Aerobiology Department, Sciensano, 1050 Brussels, Belgium • 2. BCCM/IHEM Fungal Collection, Mycology and Aerobiology Section, Sciensano, 1050 Brussels, Belgium

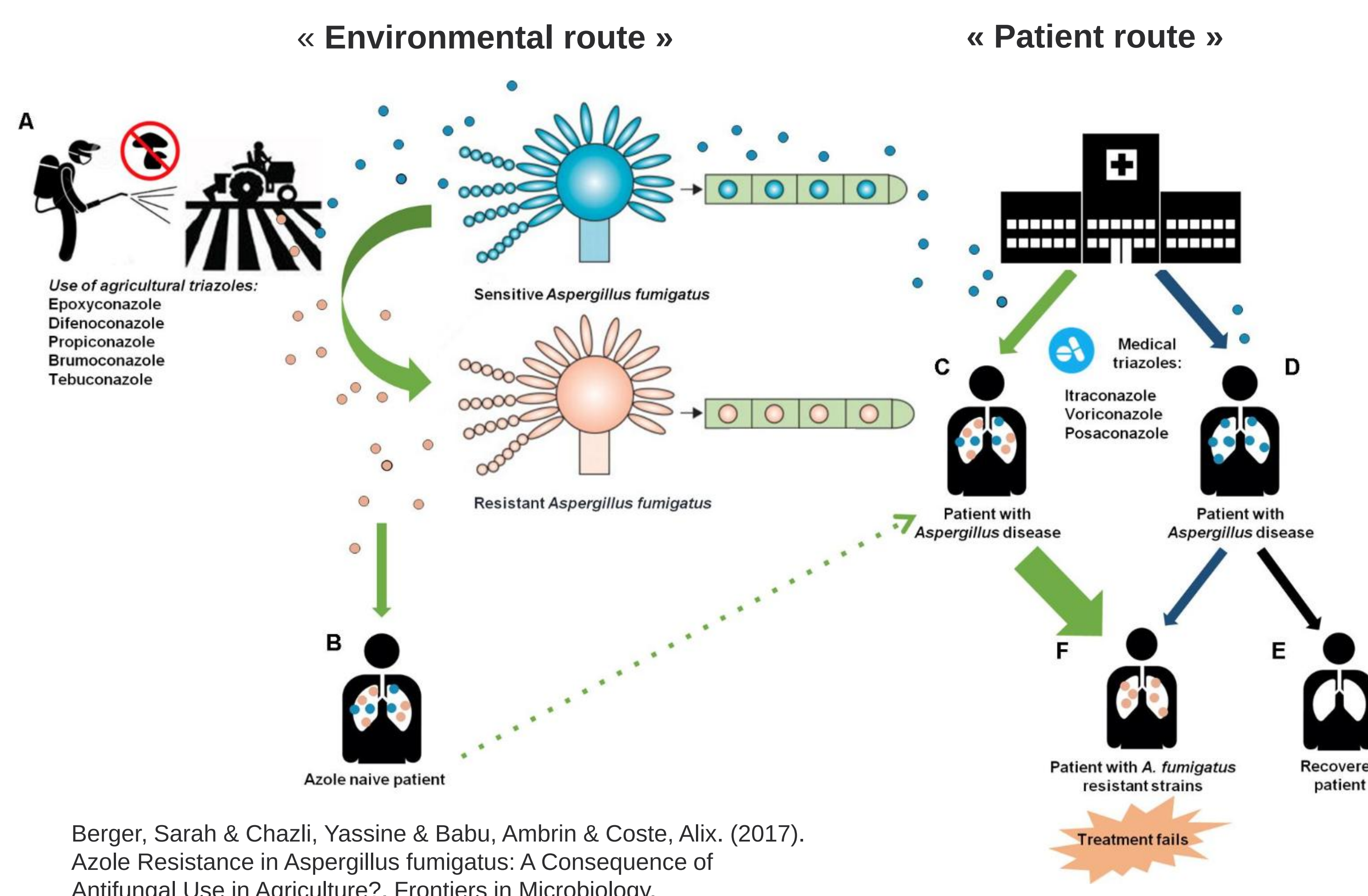
Aspergillus fumigatus can cause adverse health effects on humans, ranging from allergic reactions, allergic and chronic infections to acute invasive aspergillosis.

Triazoles are used for the prophylaxis and treatment of pathologies caused by *A. fumigatus* in clinical settings, but they are also extensively used in agriculture for pest management.

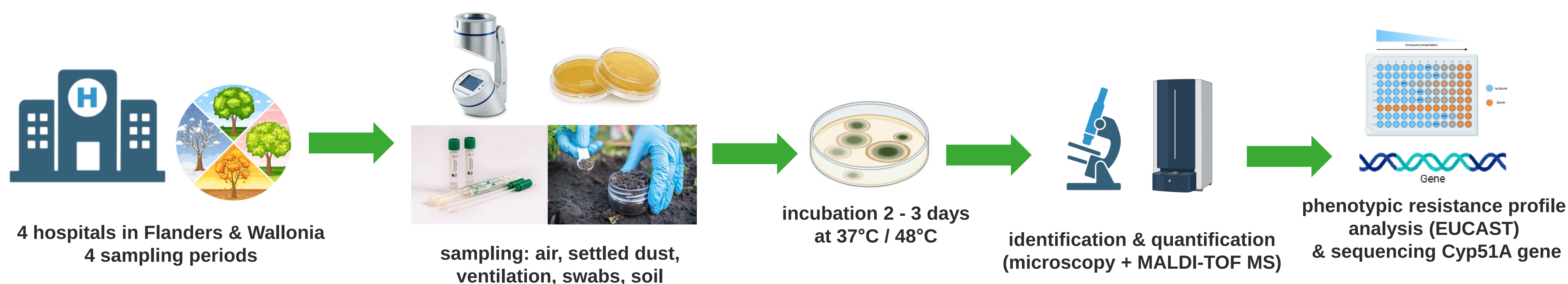
A. fumigatus triazole resistance (ARaf) development in the environment is linked to the emergence of cross-resistance of *A. fumigatus* to medical triazoles.

ARaf has become a growing concern, as highlighted by the addition of *A. fumigatus* to critical priority group of the WHO fungal priority pathogens list.

Aim of the study: asses if transmission of (azole-resistant) *A. fumigatus* from the outdoor environment to the indoor environment occurs in a hospital setting

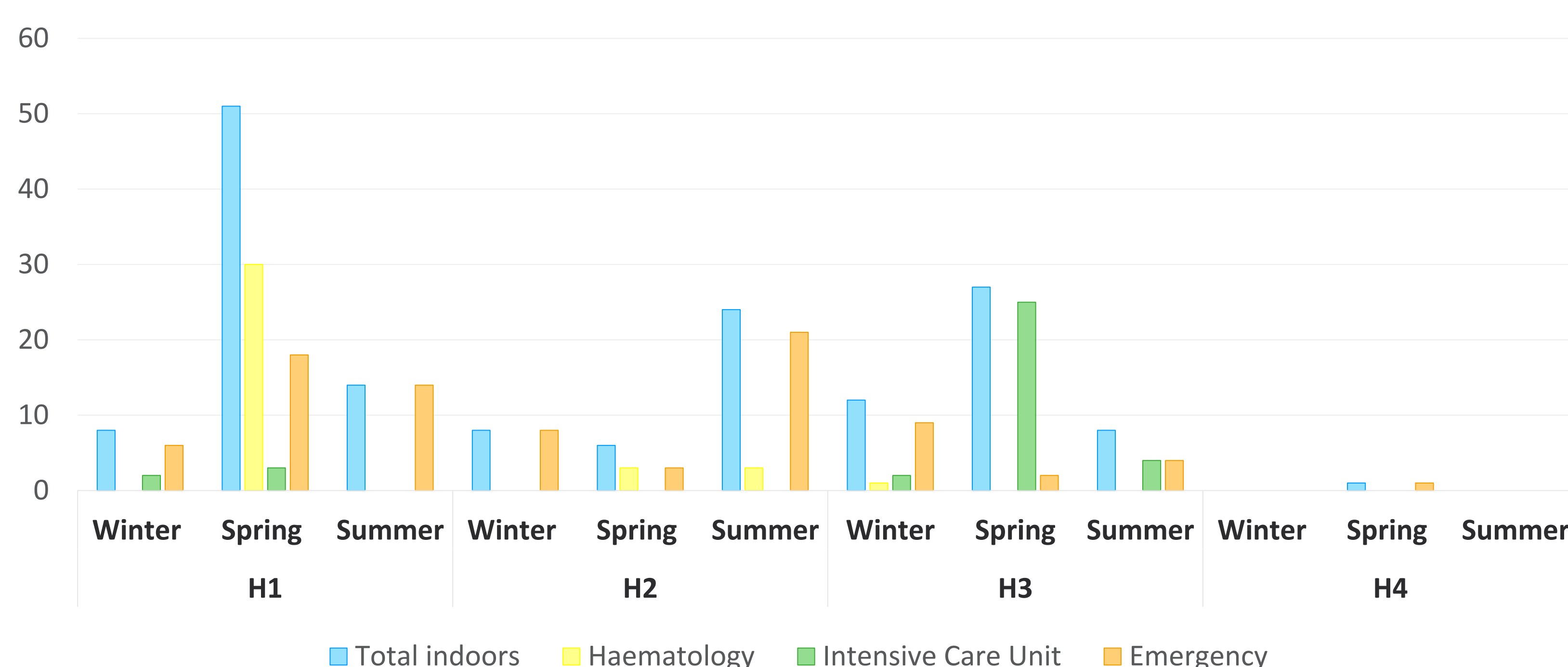


METHOD

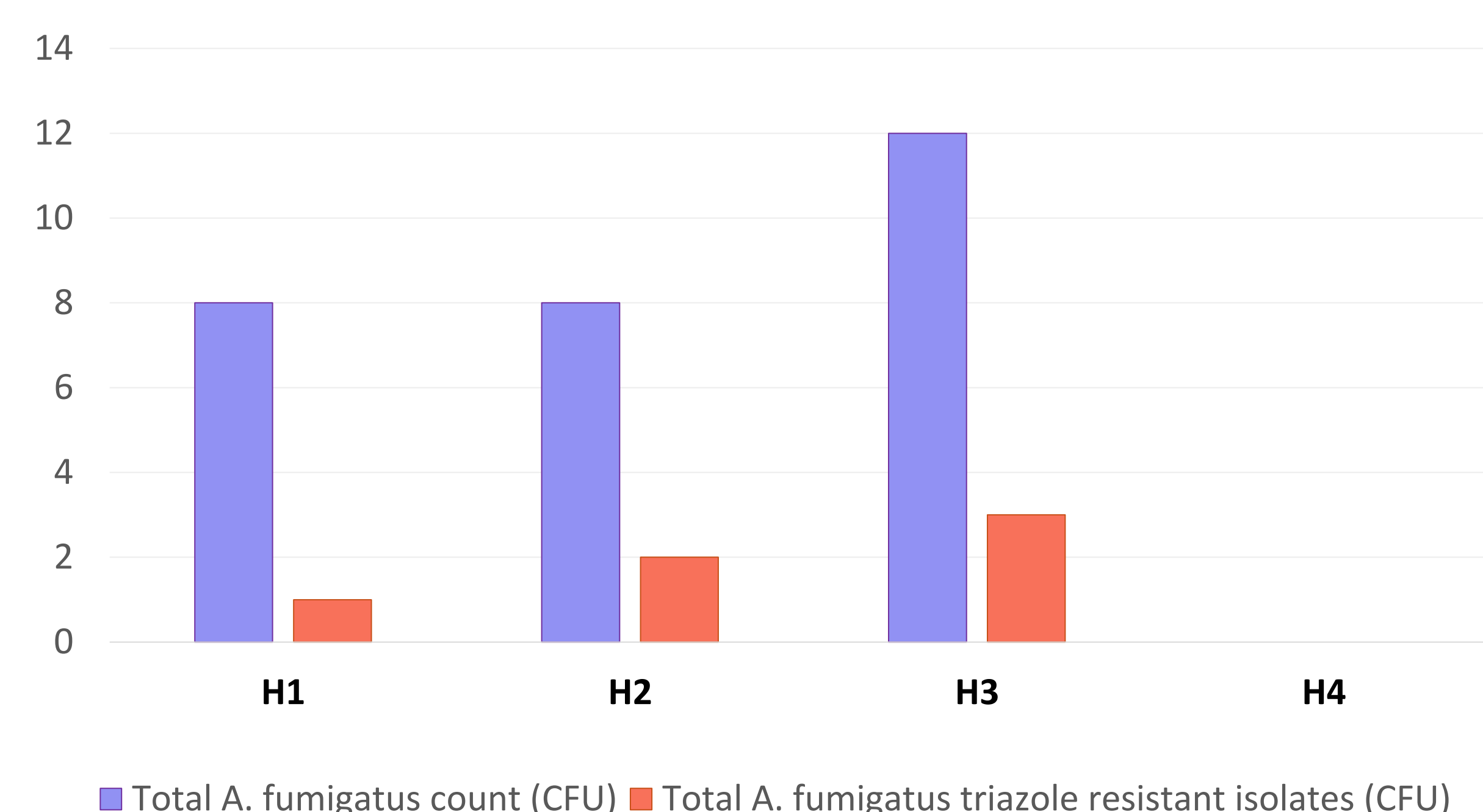


PRELIMINARY RESULTS

Aspergillus fumigatus counts (CFU) per ward and per sampling period



Aspergillus fumigatus triazole resistance in hospitals: winter sampling



PERSPECTIVES

- The project is ongoing: data of azole resistance for spring, summer and autumn samplings and data of countings for autumn sampling still need to be completed
- Factors that will be evaluated:
 - Environmental surroundings of the hospitals
 - Presence of construction sites inside and near the hospitals
 - Policies of the hospitals regarding hospital hygiene and prevention of nosocomial infections
 - Cleaning schedules and ventilation maintenance
- This project could help hospitals in their fight against hospital-acquired aspergillosis